

## Supplementary material

### Genetic Programming Hyper-Heuristics for the Job Shop Scheduling Problem with Interval Durations: Robustness-Aware Interpretable Rules that Generalize Across Instance Sizes

Sections, tables, figures and equations without the prefix S refer to the article.

## S1 Interval handling and the decoder

Two aspects of the comparison with the constructive baselines are examined here: how each baseline reduces an interval to a comparable quantity, and how much of the margin of the evolved rules belongs to the priority rule rather than to the schedule generation scheme.

The baselines rank candidates by the lexicographic order of Eq. (8) (Section 5.2). To check that this choice does not handicap them, each interval-valued rule was re-run ranking by a single number drawn from the same interval: its lower endpoint, its midpoint or its upper endpoint. Over the 70 instances SPT spans 633.1 to 642.5 across those three conventions, LPT 703.1 to 704.4, MWKR 62.9 to 64.6 and EST 42.7 to 45.1. The lexicographic values of Table 4 are better than all three for SPT and EST, and within 1.9 points of the best for LPT and MWKR. For EST and MWKR, the rules that account for the state of the shop, no convention moves the result by more than 2.4 points, and for SPT and LPT, whose levels exceed 600, by at most 9.4; the best plain rule and the evolved one are 23.3 points apart. How the intervals are collapsed is therefore not what the comparison measures.

The decoder is a more substantial question, since the strongest baseline, G&T-MWKR, does not use the same one: it restricts each decision to Giffler and Thompson’s conflict set and so produces active schedules, whereas the evolved rules choose from the whole eligible set and produce semi-active ones. Running the evolved rules as the tie-break *inside* the conflict set separates the two effects. Table S1 gives the four combinations. Holding the decoder fixed, the evolved rules outperform MWKR under either one: 18.99 against 64.7 in the semi-active decoder and 23.18 against 29.5 in G&T’s. Holding the rule fixed, each prefers a different decoder: the evolved rules lose 4.2 points when moved into the conflict set, while MWKR gains 35.2 by entering it. Each method in Table 4 is thus reported with the decoder that suits it, and the margin survives the other choice.

Table S1: RE (%) over the 70 instances for the two decoders and the two priority rules, one deterministic pass. The GP entries are the mean over the 30 evolved rules.

| Priority rule       | Semi-active             | Giffler–Thompson |
|---------------------|-------------------------|------------------|
| MWKR                | 64.7                    | 29.5             |
| Evolved, mean of 30 | <b>18.99</b> $\pm$ 1.33 | 23.18 $\pm$ 7.47 |
| Evolved, featured   | 17.71                   | 18.15            |

Two details of the table deserve mention. The featured rule is the one case where the decoder barely matters, 17.71 against 18.15, a difference of 0.44 points that a paired test over the 70 instances does not separate from zero ( $p = 0.32$ ); the loss is concentrated in the rest of the arm, whose standard deviation grows from 1.33 to 7.47 because a few rules deteriorate markedly once their candidate set is restricted. This is what should be expected of rules evolved against a particular generation scheme: the conflict set removes candidates their fitness was never measured on. And SPT, which is the worst entry of Table 4 at 627.2, reaches 70.6 inside the conflict set, as G&T-SPT, which is the same phenomenon seen from the other side.

## S2 Robustness under executional uncertainty: further analyses

This section extends the robustness comparison of Section 7.6 of the article, whose Table 8 the paragraphs below refer to. It gives the paired tests behind that comparison and the per-instance distributions under widened intervals, removes the normalization of  $\bar{\varepsilon}$ , repeats the comparison at the level of the 30 rules of each arm, changes the law of the realizations, and describes the worst 5% of executions.

The paired Wilcoxon tests behind the comparisons of the article, on  $\bar{\varepsilon}$  at the nominal width, are the following.

Under the makespan objective the width terminals make no difference to robustness either (Table 8): the featured rule and its ablated counterpart are statistically indistinguishable in  $\bar{\varepsilon}$  (5.70 against 5.75 at the nominal width; paired Wilcoxon over the 70 instances,  $z = -0.17$ ). The rule is more robust than G&T-MWKR (6.29,  $z = -4.42$ ,  $p < 0.001$ ,  $|r| = 0.61$ ) but not than EST (5.66,  $z = 0.42$ ), although it improves on EST by 24.6 points of RE; in absolute terms, which removes EST’s larger expected makespan from the denominator, the rule deviates by 12.30 time units against EST’s 14.64 ( $z = -5.89$ ,  $p < 0.001$ ,  $|r| = 0.81$ ). A low deviation does not require a low makespan, so robustness must be optimized for directly.

The robust fitness achieves this. The narrowest rule of the  $\lambda=1$  arm reduces  $\bar{\varepsilon}$  from 5.70 to 4.52 ( $z = -6.40$ ,  $|r| = 0.88$  against the makespan rule), and the reduction requires the width terminals: under the same fitness, its ablated counterpart reaches 5.54 ( $z = -6.25$ ,  $|r| = 0.86$ ). At  $\lambda=4$  the deviation falls to 4.21. In absolute terms the reduction is 16.2%, from 12.30 to 10.31 time units ( $z = -5.55$ ,  $p < 0.001$ ,  $|r| = 0.76$ ), so it does not come from the larger expected makespan in the denominator. The narrower predicted intervals of Section 7.5 thus correspond to smaller realized deviations, at the cost in expected makespan reported in the RE column: 24.95 and 28.27 against 17.71.

Methods can also be compared at similar levels of RE, which separates the robustness gain from the expected makespan given up for it. The robust  $\lambda=4$  rule and G&T-MWKR predict almost the same makespan, 2454 against 2482 time units, which is 28.27 against 29.50 points of RE. Their realized deviations differ markedly: 9.78 against 14.94 time units, a reduction of 34.5%, smaller on 69 of the 70 instances ( $z = -7.26$ ,  $p < 0.001$ ,  $|r| = 1.00$ ). At matched a-priori quality the robustness gain is therefore neither a denominator effect nor a makespan concession.

The comparison with EST reverses once the normalization is removed.  $\bar{\varepsilon}$  divides by  $E[\mathbf{C}_{\max}]$ , and EST’s predicted makespan is 21% larger than the rule’s (2732 against 2259 time units on average), so the same absolute error reads as a smaller relative one. In absolute terms the rule deviates by 12.30 time units against EST’s 14.64, and the difference is significant ( $z = -5.89$ ,  $p < 0.001$ ,  $|r| = 0.81$ ). The ordering against G&T-MWKR is unchanged and widens (12.30 against 14.94,  $z = -6.83$ ,  $|r| = 0.94$ ), and the makespan arm and its ablation remain indistinguishable (12.30 against 12.25,  $z = 0.61$ ,  $p = 0.55$ ). The normalization therefore conceals an advantage of the rule rather than creating one.

The reduction of  $\bar{\varepsilon}$  that the robust fitness achieves in the article, from 5.70 to 4.52 for the narrowest rule of the  $\lambda=1$  arm, is partly due to the denominator, and the absolute deviation quantifies it. The robust  $\lambda=1$  arm carries an  $E[\mathbf{C}_{\max}]$  that is 6.1% larger than the makespan arm’s, so where  $\bar{\varepsilon}$  falls by 20.7% the absolute deviation falls by 16.2%, from 12.30 to 10.31 time units; at  $\lambda=4$  it reaches 9.78. Roughly a fifth of the apparent gain is therefore arithmetic and the rest is not: the paired contrasts are also significant on the absolute measure, by  $-1.99$  time units against the makespan arm ( $z = -5.55$ ,  $p < 0.001$ ,  $|r| = 0.76$ ) and by  $-1.78$  against its own ablation ( $z = -5.17$ ,  $|r| = 0.71$ ).

The comparisons above select one representative rule per arm; at the level of the method the ordering is the same. Recomputing  $\bar{\varepsilon}$  at the nominal width for all 30 rules of each arm and pairing by seed, the makespan arm averages  $5.67 \pm 0.16$  against  $5.77 \pm 0.14$  for its ablation ( $z = -2.58$ ,  $p < 0.05$  after Holm’s correction over the three arm-level tests,  $|r| = 0.54$ ), the robust  $\lambda=1$  arm  $5.50 \pm 0.30$  against  $5.68 \pm 0.11$  for its own ( $z = -3.26$ ,  $p < 0.01$ ,  $|r| = 0.68$ ), and

the robust arm improves on the makespan arm with the same terminals ( $z = -2.46$ ,  $p < 0.05$ ,  $|r| = 0.51$ ).

Two things follow. The width terminals do reduce  $\bar{\varepsilon}$  slightly even under the makespan objective, a method-level effect too small to be detected in the single featured pair; and the arm-level reductions are modest, three percent or less, against the fifth separating the featured rules, so selecting the narrowest rule of the robust arm amplifies an effect that at the method level is consistent but small.

The estimate of the article draws each realization uniformly on its interval, which is one choice among several. Repeating the measurement with a symmetric triangular law, with a law skewed towards the upper endpoint, and with the single worst-case realization in which every duration takes its upper bound, leaves the ordering and every significant contrast intact. Averaged over the 70 instances, the robust  $\lambda=1$  arm improves on the makespan arm by 1.22 time units under the triangular law, 12.16 under the skewed one and 29.01 in the worst case, all with  $p < 0.001$ ; the makespan arm and its ablation remain indistinguishable under all three. The deviations themselves grow with the mass the law places near the upper endpoint, from 8.95 time units for the rule under the triangular law to 57.15 under the skewed one and 138.35 in the worst case, but the comparison between methods does not depend on which law is used.

The measures above are averages over realizations, whereas a planner exposed to late completions is concerned with the worst ones. The conditional value-at-risk of Section 5.4, computed on the same  $K=1000$  realizations, describes the 5% worst executions of each schedule and separates two questions. The first is how far those executions exceed the prediction. Under the uniform law the featured rule overruns its expected makespan by 33.46 time units on average over its worst 5% of executions, against 38.87 for G&T-MWKR and 40.81 for EST ( $z = -5.04$  and  $z = -5.61$ ,  $p < 0.001$ ), and its ablated counterpart is indistinguishable (33.55,  $z = -0.05$ ). The robust objective reduces the tail overrun in about the same proportion as the mean deviation: to 27.46 at  $\lambda=1$ , 17.9% less than the makespan rule ( $z = -5.14$ ,  $p < 0.001$ ,  $|r| = 0.71$ ) and 5.68 less than its own ablation ( $z = -5.32$ ,  $|r| = 0.73$ ), and to 26.43 at  $\lambda=4$ . At matched a-priori quality, the robust  $\lambda=4$  rule overruns by 26.43 against 38.87 for G&T-MWKR, 32.0% less ( $z = -7.00$ ,  $|r| = 0.96$ ). The second question is how long the worst executions are in absolute terms, and there the ordering follows expected makespan rather than robustness. Expressed as RE, the tail of the featured rule lies at 19.53% and those of the robust rules at 26.46% and 29.73%, higher on 68 and on 70 of the 70 instances, respectively. The robust objective therefore makes the prediction more reliable in the tail as well as on average, but it does not shorten the worst executions, whose length follows the expected makespan given up for that reliability. Against G&T-MWKR, whose expected makespan is close to its own, the  $\lambda=4$  rule has a slightly shorter tail (29.73 against 31.59,  $z = -2.41$ ,  $p = 0.016$ ,  $|r| = 0.33$ ). All of these contrasts keep their sign under the triangular and the skewed laws, under which the tail overrun at  $\lambda=1$  falls by 3.23 and 14.89 time units against the makespan rule, and all keep their significance except the shorter tail of the  $\lambda=4$  rule against G&T-MWKR, which is marginal under the triangular law ( $p = 0.044$ ).

Figure S1 shows the per-instance  $\bar{\varepsilon}$  distributions. The ordering is stable as uncertainty grows, and the advantage of the robust rule widens with it: at +40% its deviation is 6.08 against 7.57 for the makespan rule and 8.58 for G&T-MWKR.

### S3 Asymmetric intervals

Every instance used so far carries an interval symmetric about its crisp value, so what the width terminals contribute could belong to the generator rather than to the problem. The four arms of Sections 7.4 and 7.5 were therefore re-evolved on the asymmetric counterparts of TA11–TA14 described in Section 5.1, with the same configuration and fifteen seeds per arm rather than thirty, a reduction imposed by the computational cost. Table S2 reports them over all 70 asymmetric instances, with the executed deviation estimated from 400 realizations of the fixed processing

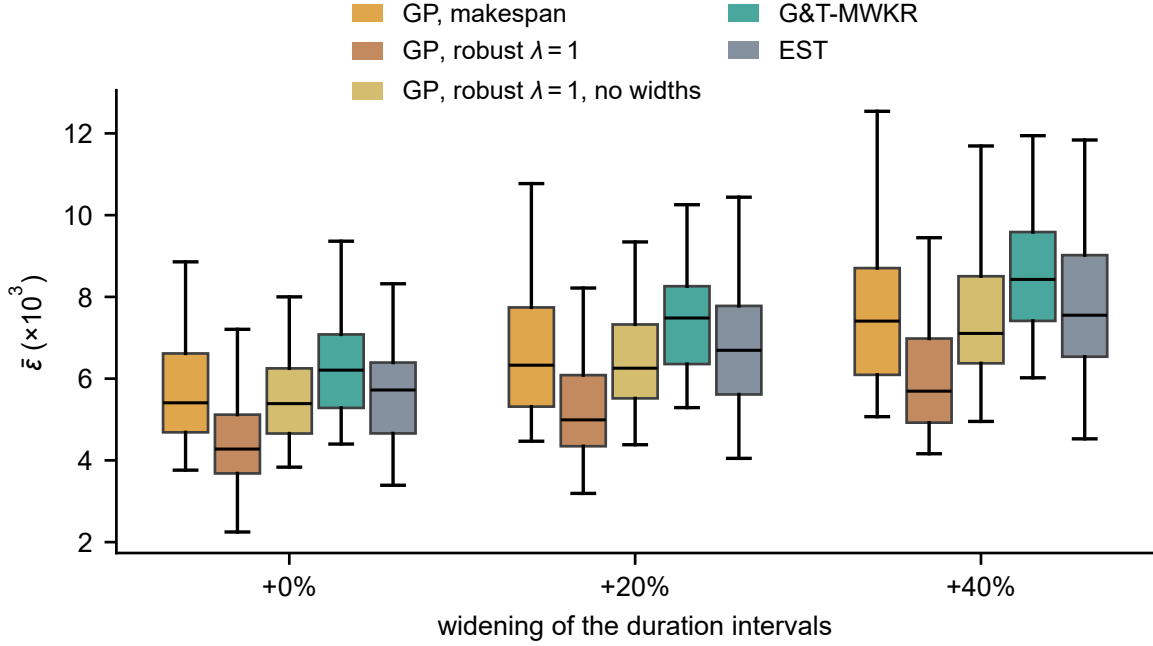


Figure S1: Per-instance  $\bar{\varepsilon}$  by method, with the duration intervals at their nominal widths and widened by +20% and +40%. The robust  $\lambda=1$  entries are the narrowest rule of that arm and its counterpart evolved without the width terminals, as in Table 8.

order. Levels of RE do not compare across the two benchmarks, since the right skew raises the expected makespan while the crisp lower bound does not move and every arm sits some six points higher; only the contrast between arms within a benchmark is meaningful.

Table S2: The four arms re-evolved on asymmetric intervals and evaluated over the 70 asymmetric instances: mean  $\pm$  sd across 15 seeds of RE, of the relative width of the predicted makespan interval, and of the absolute deviation of the executed makespan.

| Arm                           | RE (%)           | Width (%)        | $ \Delta $       |
|-------------------------------|------------------|------------------|------------------|
| Makespan, full                | $25.08 \pm 1.42$ | $13.23 \pm 0.72$ | $13.36 \pm 0.64$ |
| Makespan, no width            | $23.90 \pm 1.18$ | $13.08 \pm 0.20$ | $13.06 \pm 0.26$ |
| Robust $\lambda=1$ , full     | $25.93 \pm 1.18$ | $12.67 \pm 0.20$ | $12.75 \pm 0.27$ |
| Robust $\lambda=1$ , no width | $23.89 \pm 1.06$ | $12.97 \pm 0.18$ | $13.03 \pm 0.16$ |

The main result is reproduced. Under the robust objective the width terminals narrow the predicted interval from 12.97 to 12.67 ( $z = -3.24$ ,  $|r| = 0.95$ ) and the executed deviation from 13.03 to 12.75 ( $z = -2.90$ ,  $|r| = 0.85$ ), and the robust arm improves on the makespan arm carrying the same terminals by 0.55 points of width and 0.61 time units of deviation, the latter for every one of the fifteen seeds ( $z = -3.41$ ,  $|r| = 1.00$ ). These five contrasts survive Holm's correction over the nine tests of the table, the largest adjusted  $p$  among them being 0.0101. The robustness that the width terminals make attainable, and the objective under which they attain it, therefore do not depend on the symmetric generation scheme.

Under the makespan objective the results are weaker than on the symmetric benchmark. There, the width terminals made the predicted interval slightly but significantly narrower even though that objective does not reward it (Section 7.4); here they do not (13.23 against 13.08,  $p = 0.89$ ), nor do they reduce the executed deviation (13.36 against 13.06,  $p = 0.11$ ), and they cost 1.18 points of RE, a difference that does not survive the correction ( $p = 0.10$  adjusted).

The negative result for expected makespan holds on both benchmarks, and the remaining results are consistent with the boundary of Section 7.5: where the objective does not price width, the smaller terminal set is the better choice.

## S4 Per-instance results

Table S3 lists, for each of the 70 interval Taillard instances, the crisp lower bound and the single-pass RE (%) of the evolved rule and of the two strongest baselines: EST, the best of the plain dispatching rules, and G&T-MWKR, the best baseline overall.

Table S3: Per-instance RE (%) of the evolved rule (GP) and of the two strongest baselines, EST and G&T-MWKR (single deterministic pass). LB is the crisp Taillard lower bound.

| Inst. | LB   | EST  | G&T  | GP   | Inst. | LB   | EST  | G&T  | GP   |
|-------|------|------|------|------|-------|------|------|------|------|
| TA1   | 1231 | 49.9 | 26.1 | 19.3 | TA36  | 1819 | 41.1 | 42.8 | 23.8 |
| TA2   | 1244 | 24.9 | 27.8 | 14.2 | TA37  | 1771 | 43.4 | 35.6 | 18.4 |
| TA3   | 1218 | 33.0 | 33.2 | 14.1 | TA38  | 1673 | 45.0 | 28.0 | 20.1 |
| TA4   | 1175 | 32.0 | 27.7 | 13.0 | TA39  | 1795 | 46.6 | 28.9 | 22.4 |
| TA5   | 1224 | 27.2 | 33.0 | 14.7 | TA40  | 1651 | 46.9 | 42.3 | 26.1 |
| TA6   | 1238 | 26.8 | 18.2 | 13.7 | TA41  | 1906 | 44.6 | 45.0 | 33.4 |
| TA7   | 1227 | 30.7 | 26.1 | 17.7 | TA42  | 1884 | 71.1 | 38.1 | 24.3 |
| TA8   | 1217 | 23.7 | 25.0 | 14.8 | TA43  | 1809 | 68.7 | 39.8 | 22.3 |
| TA9   | 1274 | 37.7 | 28.2 | 20.0 | TA44  | 1948 | 52.1 | 34.7 | 25.8 |
| TA10  | 1241 | 37.0 | 22.1 | 15.4 | TA45  | 1997 | 50.2 | 33.7 | 14.9 |
| TA11  | 1357 | 56.2 | 30.7 | 17.5 | TA46  | 1957 | 60.8 | 31.6 | 22.4 |
| TA12  | 1367 | 48.8 | 34.6 | 17.6 | TA47  | 1807 | 46.8 | 30.1 | 25.6 |
| TA13  | 1342 | 60.2 | 26.9 | 13.3 | TA48  | 1912 | 59.1 | 33.7 | 21.1 |
| TA14  | 1345 | 39.5 | 29.7 | 13.2 | TA49  | 1931 | 51.0 | 35.3 | 20.1 |
| TA15  | 1339 | 48.1 | 30.6 | 16.7 | TA50  | 1833 | 61.4 | 46.2 | 31.3 |
| TA16  | 1360 | 45.2 | 34.0 | 15.6 | TA51  | 2760 | 33.8 | 34.7 | 23.2 |
| TA17  | 1462 | 58.9 | 17.7 | 13.9 | TA52  | 2756 | 38.0 | 26.5 | 12.7 |
| TA18  | 1377 | 43.8 | 31.8 | 23.6 | TA53  | 2717 | 34.1 | 18.3 | 11.8 |
| TA19  | 1332 | 45.8 | 29.3 | 20.0 | TA54  | 2839 | 20.9 | 16.1 | 8.2  |
| TA20  | 1348 | 45.7 | 24.1 | 15.1 | TA55  | 2679 | 31.7 | 29.2 | 15.5 |
| TA21  | 1642 | 33.4 | 28.0 | 20.2 | TA56  | 2781 | 34.8 | 19.8 | 10.9 |
| TA22  | 1561 | 44.2 | 30.8 | 17.4 | TA57  | 2943 | 32.9 | 20.0 | 9.6  |
| TA23  | 1518 | 42.8 | 35.3 | 17.8 | TA58  | 2885 | 37.4 | 26.3 | 13.6 |
| TA24  | 1644 | 56.3 | 23.8 | 14.1 | TA59  | 2655 | 26.6 | 28.2 | 16.7 |
| TA25  | 1558 | 45.8 | 33.8 | 16.6 | TA60  | 2723 | 41.7 | 19.8 | 15.5 |
| TA26  | 1591 | 42.8 | 31.2 | 26.9 | TA61  | 2868 | 45.7 | 25.4 | 15.4 |
| TA27  | 1652 | 50.0 | 31.1 | 21.4 | TA62  | 2869 | 43.0 | 27.6 | 17.2 |
| TA28  | 1603 | 36.8 | 26.8 | 12.4 | TA63  | 2755 | 38.8 | 23.8 | 13.4 |
| TA29  | 1583 | 28.8 | 32.0 | 17.2 | TA64  | 2702 | 38.2 | 28.4 | 12.2 |
| TA30  | 1528 | 34.2 | 36.1 | 17.0 | TA65  | 2725 | 42.9 | 30.0 | 20.5 |
| TA31  | 1764 | 30.7 | 34.4 | 24.5 | TA66  | 2845 | 41.8 | 28.0 | 11.9 |
| TA32  | 1774 | 42.7 | 33.3 | 19.0 | TA67  | 2825 | 31.7 | 26.1 | 15.2 |
| TA33  | 1788 | 51.3 | 34.8 | 27.5 | TA68  | 2784 | 50.1 | 19.1 | 10.6 |
| TA34  | 1828 | 45.9 | 31.8 | 20.7 | TA69  | 3071 | 35.3 | 22.8 | 12.9 |
| TA35  | 2007 | 30.3 | 24.1 | 8.4  | TA70  | 2995 | 42.9 | 23.4 | 16.7 |

## S5 The seeded genetic algorithm over time

Table S4 compares the genetic algorithm seeded with the rule’s permutation against the unseeded one at several budgets on the four instance sets of Section 6.5 of the article. The seed gives a significant advantage on every set after one second per instance, for longer the larger the instance, and none from 100 s per instance on. The tests at different budgets are not corrected for multiplicity; with 10 instances the smallest attainable two-sided  $p$  is 0.002, the value repeated in the first rows.

## S6 Ablation results

Table S5 gives the means of the arms compared seed by seed in Figure 8 of the article.

Table S4: The seeded genetic algorithm against the unseeded one: number of instances on which the seeded version is better and, in parentheses, the two-sided paired Wilcoxon  $p$ , with the instance as unit (the mean over its runs). A dash marks a budget beyond the horizon of the set.

| Seconds        | Classical (12) | 15×15 (10) | 30×15 (10) | 50×15 (10) |
|----------------|----------------|------------|------------|------------|
| 1              | 11 (0.001)     | 10 (0.002) | 10 (0.002) | 10 (0.002) |
| 3              | 8 (0.064)      | 10 (0.002) | 10 (0.002) | 10 (0.002) |
| 10             | 5 (0.970)      | 6 (0.625)  | 9 (0.004)  | 10 (0.002) |
| 30             | 5 (0.970)      | 6 (0.625)  | 5 (0.846)  | 9 (0.006)  |
| 100            | 6 (0.791)      | 5 (1.000)  | 4 (0.432)  | 8 (0.064)  |
| 300            | 6 (0.791)      | 5 (0.846)  | 5 (0.846)  | 6 (0.432)  |
| 1000           | —              | —          | 4 (0.846)  | 6 (0.625)  |
| End of horizon | 6 (0.791)      | 3 (0.625)  | 4 (0.375)  | 7 (0.432)  |

Table S5: Ablations of the interval machinery (30 independent evolutions per arm; paired rank tests between arms): the terminal ablation under two fitness objectives, and the crisp-midpoint control, evolved with scalar fitness and no interval information, tested against the interval-trained arm above it. *Width* is the relative width of the predicted makespan interval.

| Fitness objective               | Terminal set   | RE (%)                 | Width (%)             |
|---------------------------------|----------------|------------------------|-----------------------|
| Makespan $E[\mathbf{C}_{\max}]$ | full           | $18.99 \pm 1.33$       | $12.41 \pm 0.30$      |
|                                 | without widths | $18.66 \pm 1.08$       | $12.62 \pm 0.23$      |
|                                 | <i>test</i>    | $n.s. (z = -0.83)$     | $z = 3.40, p < 0.001$ |
| Midpoint control                | without widths | $18.63 \pm 0.99$       | $12.55 \pm 0.17$      |
|                                 | <i>test</i>    | $n.s. (z = -0.13)$     | $n.s. (z = -1.29)$    |
| Robust, Eq. (13)                | full           | $19.62 \pm 1.52$       | $12.04 \pm 0.75$      |
|                                 | without widths | $18.42 \pm 0.87$       | $12.47 \pm 0.20$      |
|                                 | <i>test</i>    | $z = -3.75, p < 0.001$ | $z = 3.22, p < 0.001$ |